IEEE 802.11

Integrated Communication Systems Group
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Basics

Questions to be answered:

- What is the difference between infrastructure and ad-hoc mode?
- What is the purpose of the distribution system?
- Which protocol layers does the 802.11 standard comprise?
- What are the main differences between the different PHY variants?

Characteristics of Wireless LANs

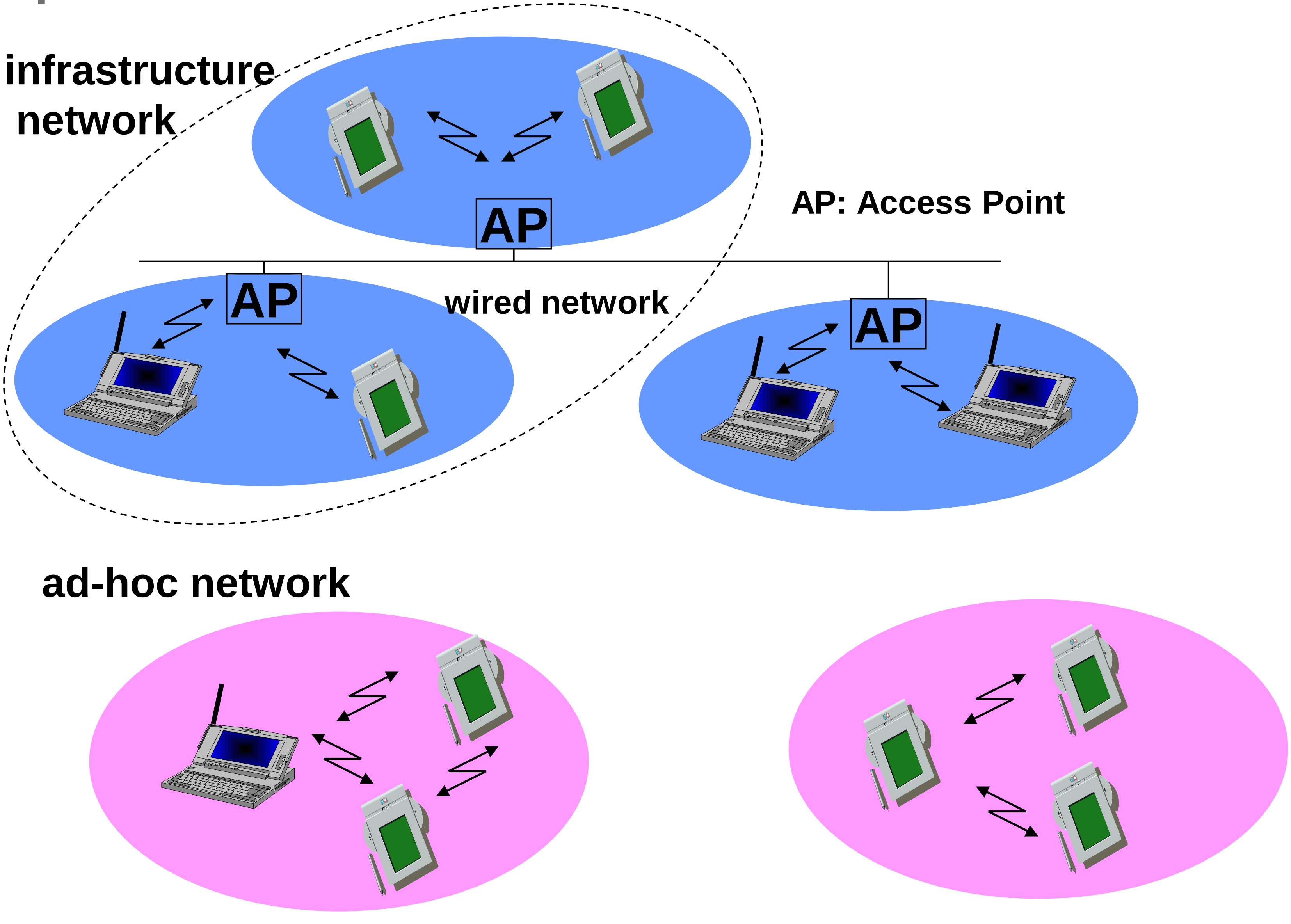
- Advantages over wired LANs
  - very flexible alternative
  - (almost) no wiring difficulties (e.g. historic buildings, firewalls)
  - ad-hoc networks without previous planning possible
  - more robust against disasters, e.g. earthquakes, fire, or users pulling a plug ...

- Disadvantages
  - shared medium
  - lower bandwidth compared to wired networks
  - possible interference may reduce bandwidth
  - no guaranteed service due to license-free spectrum
  - need to consider security issues
  - wireless products have to follow national restrictions

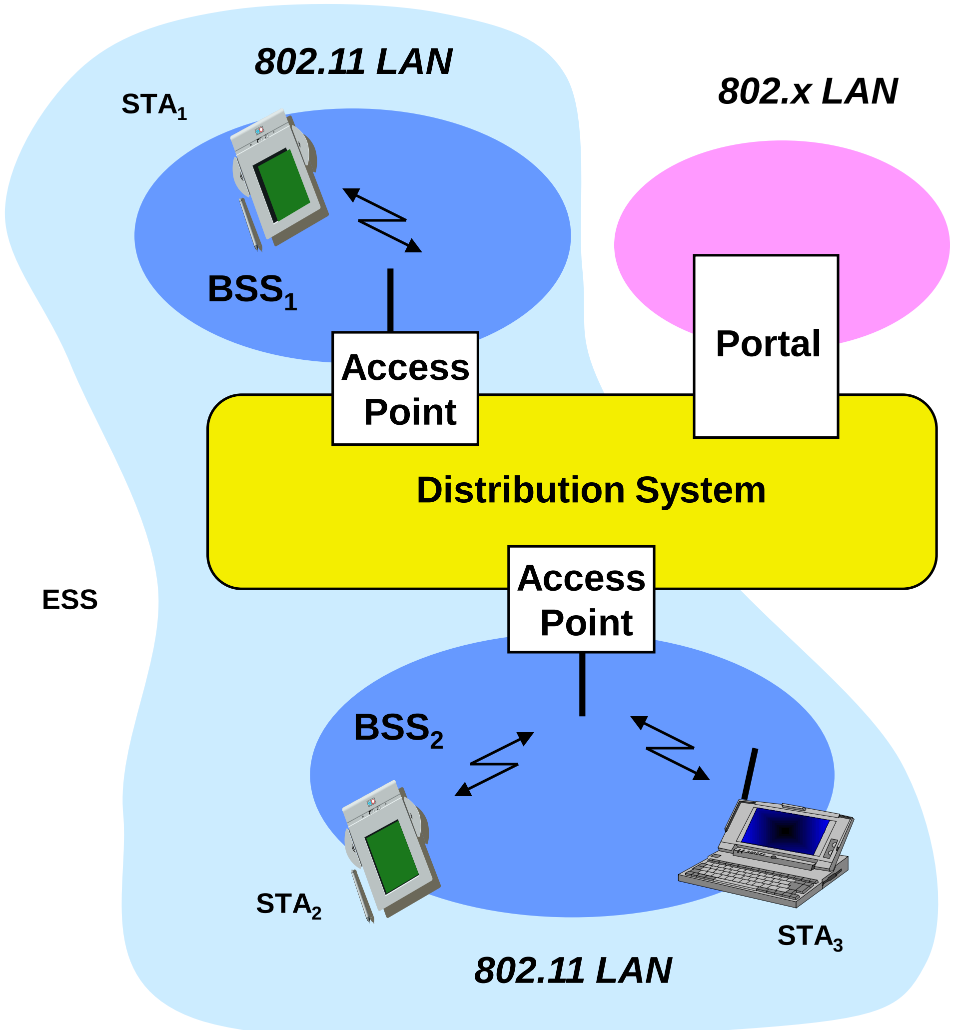
Design Goals for Wireless LANs

- Global, seamless operation
- Low power for battery use
- No special permissions or licenses needed
- Robust transmission technology
- Simplified spontaneous cooperation at meetings
- Easy to use for everyone, simple management
- Protection of investment in wired networks
- Security (no one should be able to read my data)
- Privacy (no one should be able to collect user profiles)
- Safety (low radiation)
- Transparency concerning applications and higher layer protocols, but also location awareness if necessary

Comparison: Infrastructure vs. Ad-hoc Networks

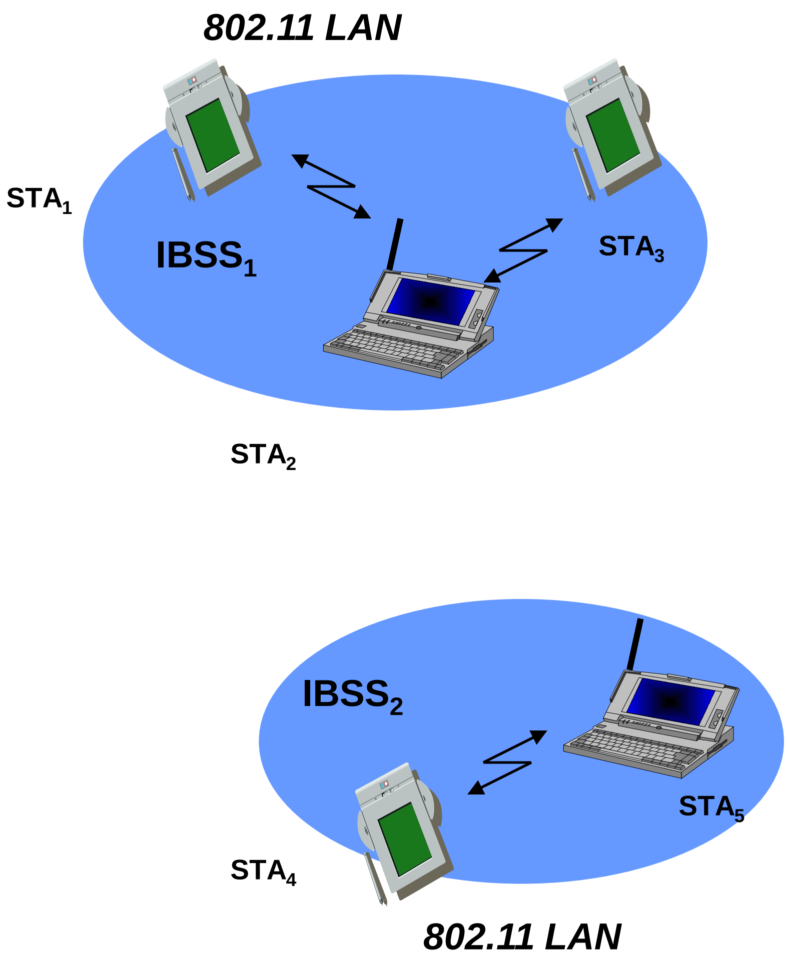


802.11: Architecture of an Infrastructure Network



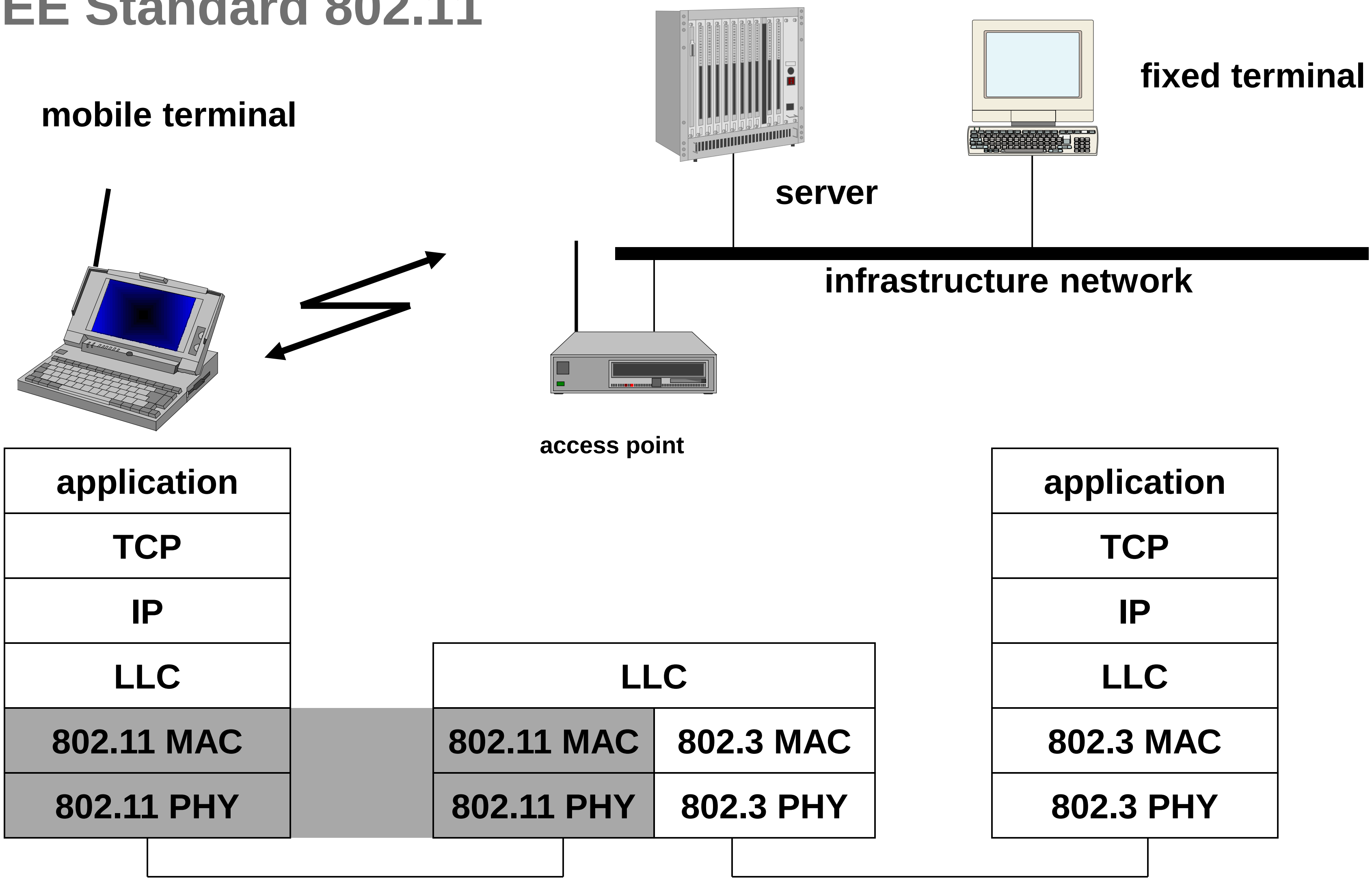
- Station (STA)
  - terminal with access mechanisms to the wireless medium and radio contact to the access point
- Basic Service Set (BSS)
  - group of stations using the same radio frequency
- Access Point
  - station integrated into the wireless LAN and the distribution system
- Portal
  - bridge to other (wired) networks
- Distribution System
  - interconnection network to form one logical network (ESS: Extended Service Set) based on several BSS

802.11: Architecture of an Ad-hoc Network



- Direct communication within a limited range
  - Station (STA):
    terminal with access mechanisms to the wireless medium
  - Independent Basic Service Set (IBSS):
    group of stations using the same radio frequency

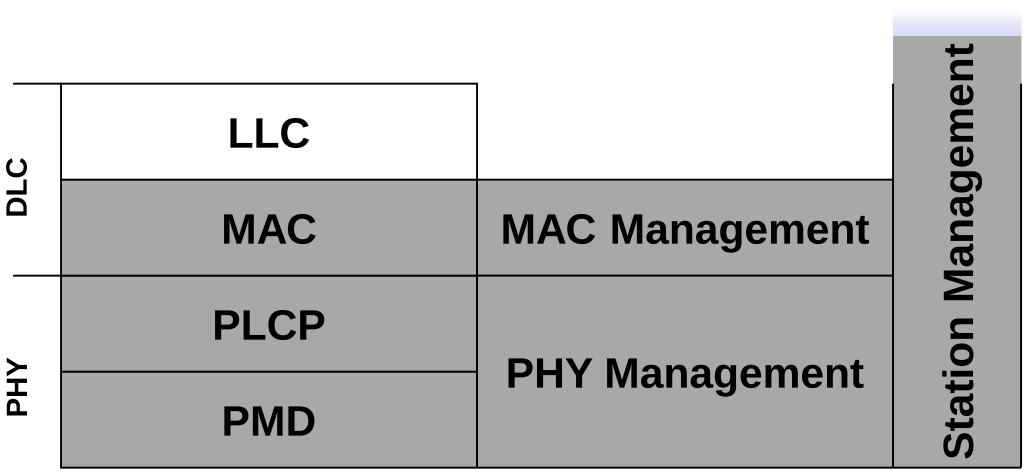
IEEE Standard 802.11



802.11 – Layers and Functions

- MAC
  - access mechanisms, fragmentation, encryption
- MAC Management
  - synchronization, roaming, Management Information Base (MIB), power management

- PLCP (Physical Layer Convergence Protocol)
  - clear channel assessment signal (carrier sense)
- PMD (Physical Medium Dependent)
  - modulation, coding
- PHY Management
  - channel selection, MIB
- Station Management
  - coordination of all management functions



802.11 – Physical Layer (original standard)

- 3 versions: 2 radio (typ. 2.4 GHz), 1 IR
  - data rates 1 or 2 Mbit/s
- FHSS (Frequency Hopping Spread Spectrum)
  - spreading, despreading, signal strength, typ. 1 Mbit/s
  - min. 2.5 frequency hops/s (USA), two-level GFSK modulation
- DSSS (Direct Sequence Spread Spectrum)
  - DBPSK modulation for 1 Mbit/s (Differential Binary Phase Shift Keying), DQPSK for 2 Mbit/s (Differential Quadrature PSK)
  - preamble and header of a frame is always transmitted with 1 Mbit/s, rest of transmission 1 or 2 Mbit/s
  - chipping sequence: +1, -1, +1, +1, -1, +1, +1, +1, -1, -1, -1 (Barker code)
  - max. radiated power 1 W (USA), 100 mW (EU), min. 1mW
- Infrared
  - 850-950 nm, diffuse light, typ. 10 m range
  - carrier detection, energy detection, synchronization

802.11 extensions – Physical Layer

v·t·e **802.11 network PHY standards** [hide]

802.11 protocol	Release date[6]	Fre-quency (GHz)	Band-width (MHz)	Stream data rate[7] (Mbit/s)	Allowable MIMO streams	Modulation	Approximate range[citation needed] Indoor (m)	Indoor (ft)	Outdoor (m)	Outdoor (ft)
802.11-1997	Jun 1997	2.4	22	1, 2	N/A	DSSS, FHSS	20	66	100	330
a	Sep 1999	5	20	6, 9, 12, 18, 24, 36, 48, 54	N/A	OFDM	35	115	120	390
		3.7[A]					—	—	5,000	16,000[A]
b	Sep 1999	2.4	22	1, 2, 5.5, 11	N/A	DSSS	35	115	140	460
g	Jun 2003	2.4	20	6, 9, 12, 18, 24, 36, 48, 54	N/A	OFDM	38	125	140	460
n	Oct 2009	2.4/5	20	7.2, 14.4, 21.7, 28.9, 43.3, 57.8, 65, 72.2 [B] (6.5, 13, 19.5, 26, 39, 52, 58.5, 65) [C]	4	OFDM	70	230	250	820[8]
			40	15, 30, 45, 60, 90, 120, 135, 150 [B] (13.5, 27, 40.5, 54, 81, 108, 121.5, 135) [C]			70	230	250	820[8]
ac	Dec 2013	5	20	7.2, 14.4, 21.7, 28.9, 43.3, 57.8, 65, 72.2, 86.7, 96.3 [B] (6.5, 13, 19.5, 26, 39, 52, 58.5, 65, 78, 86.7) [C]	8		35	115[9]		
			40	15, 30, 45, 60, 90, 120, 135, 150, 180, 200 [B] (13.5, 27, 40.5, 54, 81, 108, 121.5, 135, 162, 180) [C]			35	115[9]		
			80	32.5, 65, 97.5, 130, 195, 260, 292.5, 325, 390, 433.3 [B] (29.2, 58.5, 87.8, 117, 175.5, 234, 263.2, 292.5, 351, 390) [C]			35	115[9]		
			160	65, 130, 195, 260, 390, 520, 585, 650, 780, 866.7 [B] (58.5, 117, 175.5, 234, 351, 468, 702, 780) [C]			35	115[9]		
ad	Dec 2012	60	2,160	Up to 6,912 (6.75 Gbit/s) [10]	N/A	OFDM, single carrier, low-power single carrier	60	200	100	300
ah	Est. 2016[6]	0.9								
aj	Est. 2016[6]	45/60								
ax	Est. 2019[6]	2.4/5				MIMO-OFDM				
ay	2017	60	8000	Up to 100 (100 Gbit/s)	4	OFDM, single carrier,	60	200	1000	3000

MAC

Questions to be answered:

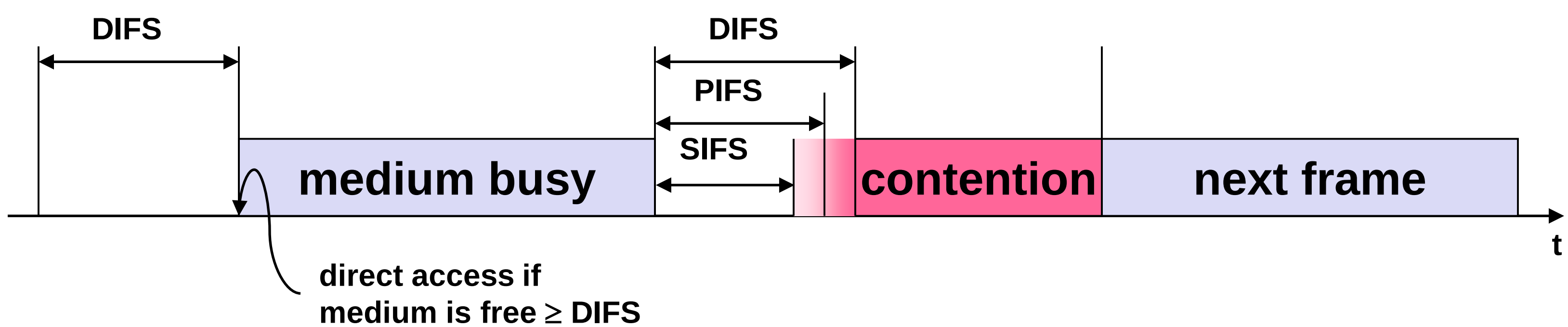
- Exponential backoff: what does it mean? What is its purpose?
- Describe DCF and PCF. What are the differences, pros and cons of each approach?
- DCF: what is the purpose of different interframe spaces?
- Assume a node in DCF mode with RTS/CTS activated wants to transmit multiple segments of data: are multiple RTS/CTS cycles required for this?
- For which messages does the SIFS apply? For what reason?
- What is the purpose of the NAV?
- How does DCF handle situations with high load? What are the strategies?

802.11 MAC Layer – DFWMAC

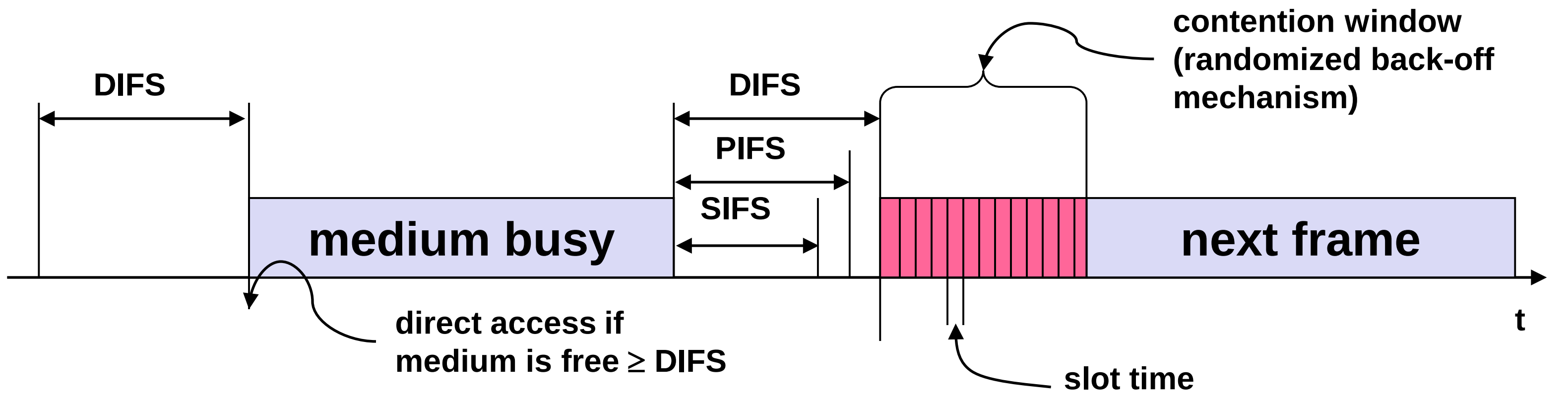
- Traffic services
  - Asynchronous Data Service (mandatory)
    - exchange of data packets based on "best-effort"
    - support of broadcast and multicast
    - implemented using DCF (Distributed Coordination Function)
  - Time-Bounded Service (optional)
    - implemented using PCF (Point Coordination Function)
- Access methods
  - DFWMAC-DCF CSMA/CA (mandatory)
    - **D**istributed **F**oundation **W**ireless **MAC**
    - collision avoidance via randomized "back-off" mechanism
    - minimum distance between consecutive packets
    - ACK packet for acknowledgements (not for broadcasts)
  - DFWMAC-DCF with RTS/CTS Extension (optional)
    - avoids hidden terminal problem
  - DFWMAC-PCF (optional)
    - access point polls terminals according to a list

802.11 MAC

- Priorities
  - defined through different Inter-Frame Spaces (IFSs)
  - no guaranteed or hard priorities
  - **SIFS** (Short Inter-Frame Spacing)
    - highest priority, for ACK, CTS, polling response
  - **PIFS** (PCF IFS)
    - medium priority, for time-bounded service using PCF
  - **DIFS** (DCF IFS)
    - lowest priority, for asynchronous data service



802.11 – CSMA/CA Access Method

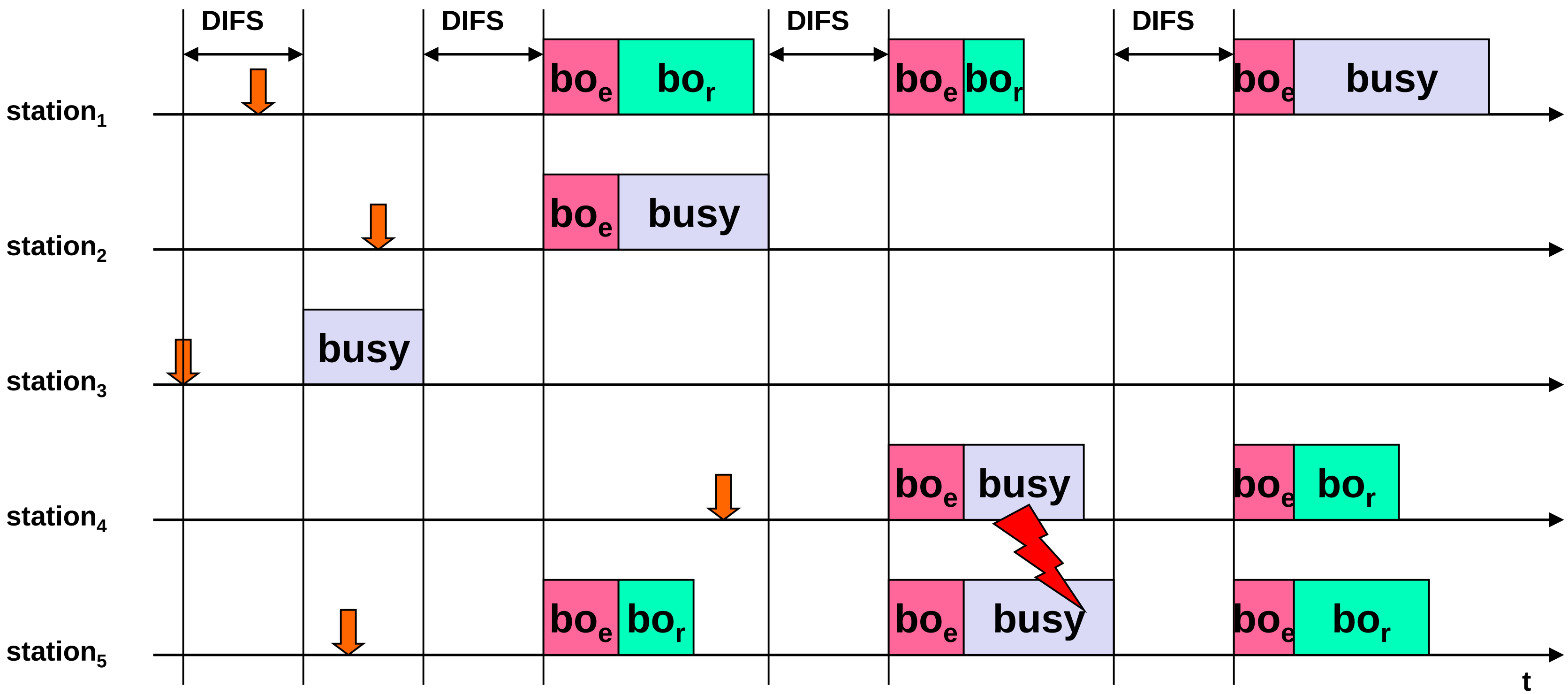


- Station ready to send starts sensing the medium (Carrier Sense based on CCA, Clear Channel Assessment)
- If the medium is free for the duration of an Inter-Frame Space (IFS), the station can start sending (IFS depends on service type)
- If the medium is busy, the station has to wait for a free IFS, then the station must additionally wait a random back-off time

  Random() = Pseudorandom integer drawn from a uniform distribution over the interval [0, CW]

  CW = an integer between CWmin and CWmax
- If another station occupies the medium during the back-off time of the station, the back-off timer stops (fairness)
- If another collision happens, the stations double CW

802.11 – Competing Stations – Simple Version



busy — medium not idle (frame, ack etc.)

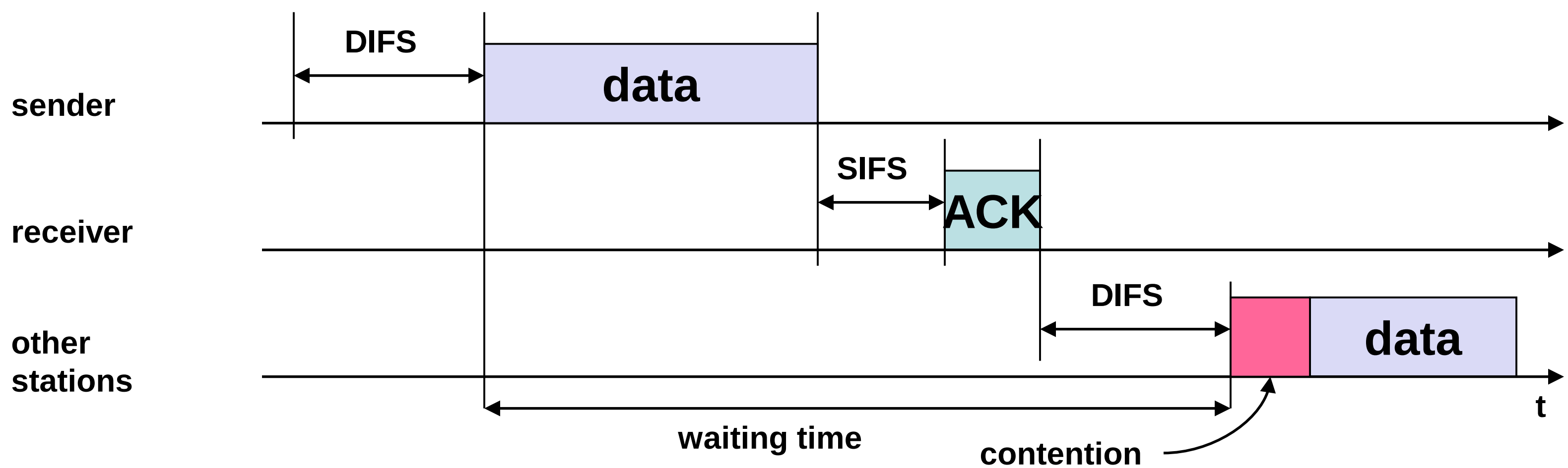
bo$_e$ — elapsed backoff time

packet arrival at MAC

bo$_r$ — residual backoff time

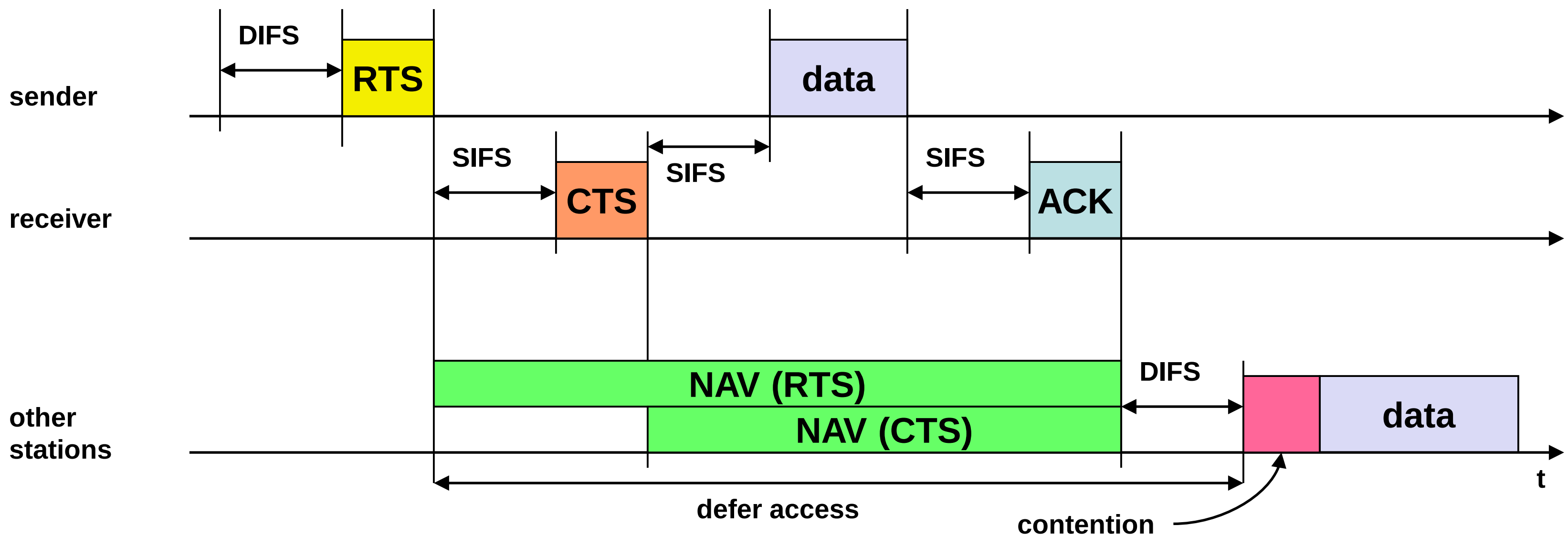
802.11 – CSMA/CA Access Method

- Sending unicast packets
  - station has to wait for DIFS before sending data
  - receivers acknowledge at once (after waiting for SIFS) if the packet was received correctly (CRC)
  - automatic retransmission of data packets in case of transmission errors



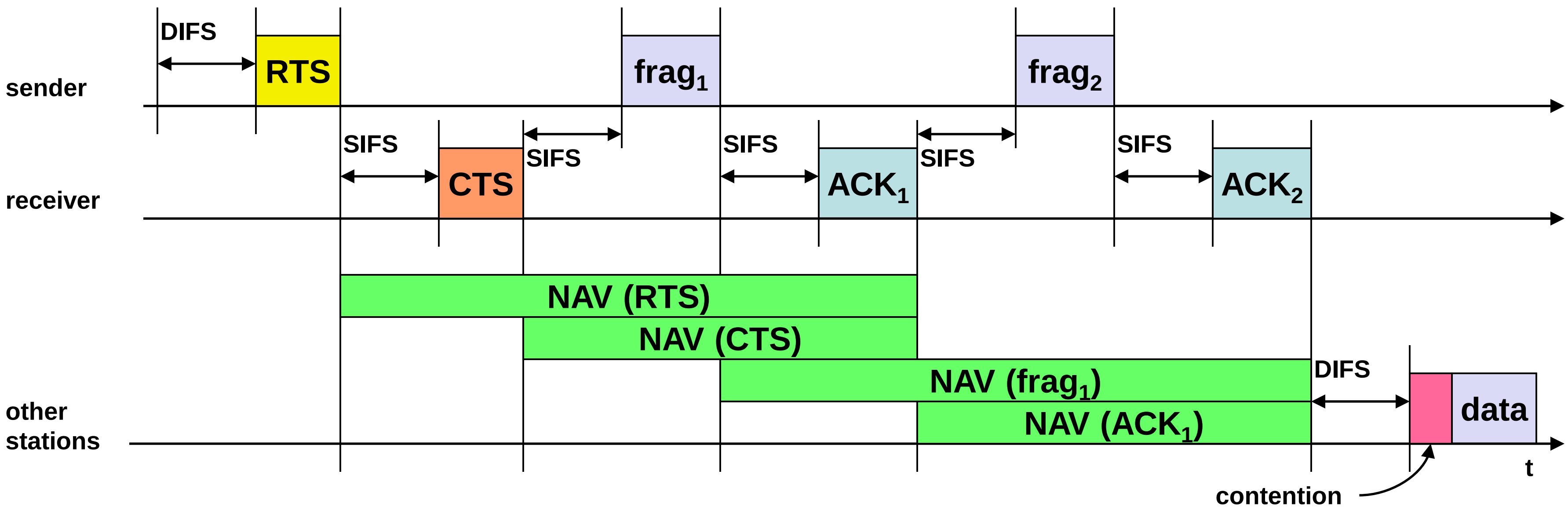
802.11 – DCF with RTS/CTS Extension

- Sending unicast packets
  - station can send RTS with reservation parameter after waiting for DIFS (reservation determines amount of time the data packet needs the medium)
  - acknowledgement via CTS after SIFS by receiver (if ready to receive)
  - sender can now send data at once, acknowledgement via ACK
  - other stations store medium reservations distributed via RTS and CTS



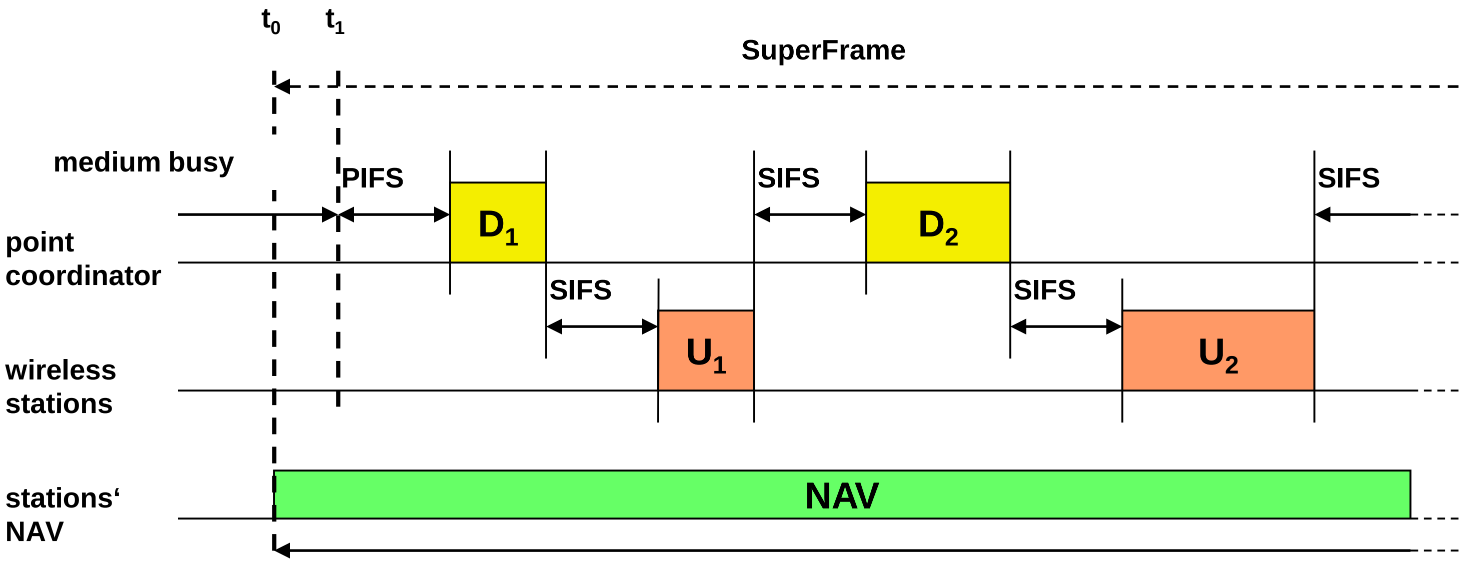
NAV: network allocation vector (implicit in RTS and CTS)

Fragmentation



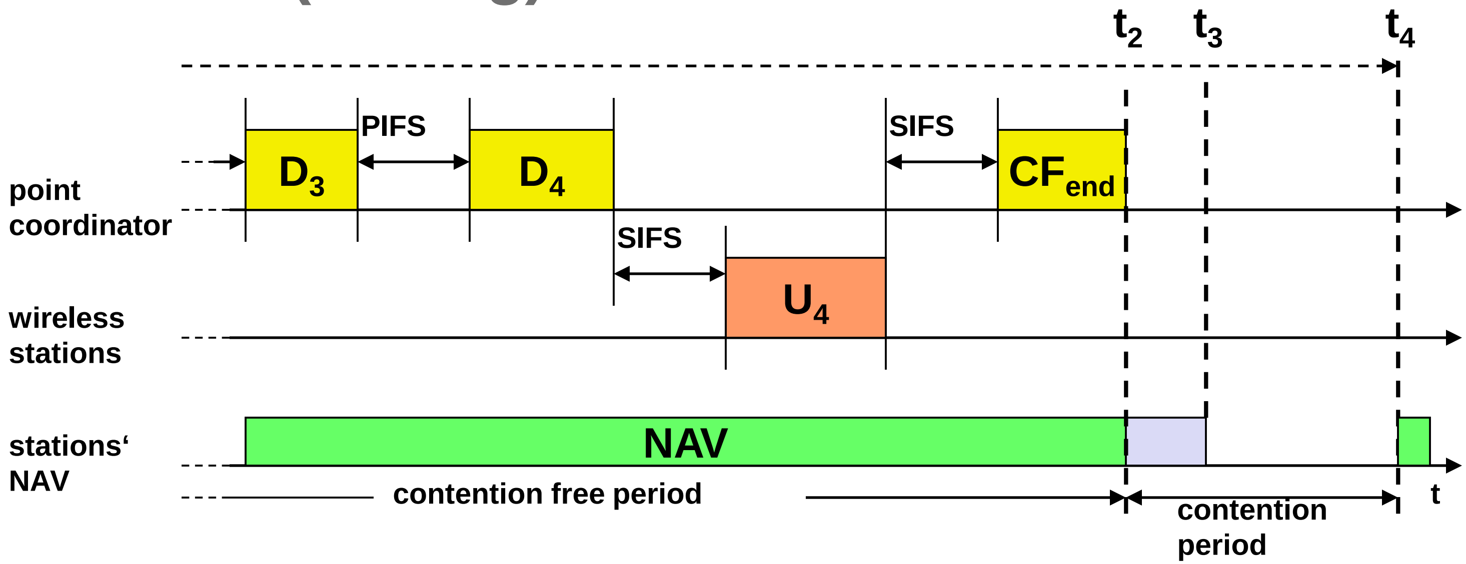
802.11 – PCF (Polling)

SuperFrame defines time span for polling of (all) wireless stations by AP (including time to reply)



D1: polling of wireless station 1
U1: station 1 responds to polling by sending its data
D2: polling of wireless station 2
U2: station 2 responds to polling by sending its data

802.11 – PCF (Polling)



D3: polling of wireless station 3
U3: no data -> no response by station 3 within SIFS
D4 (after PIFS): polling of wireless station 4
U4: station 4 responds to polling by sending its data

D1 … D4 may contain data as well

Discussion:

- unknown transmission duration of polled stations
- unpredictable beacon delays

⇒ no QoS guarantees

MAC Management

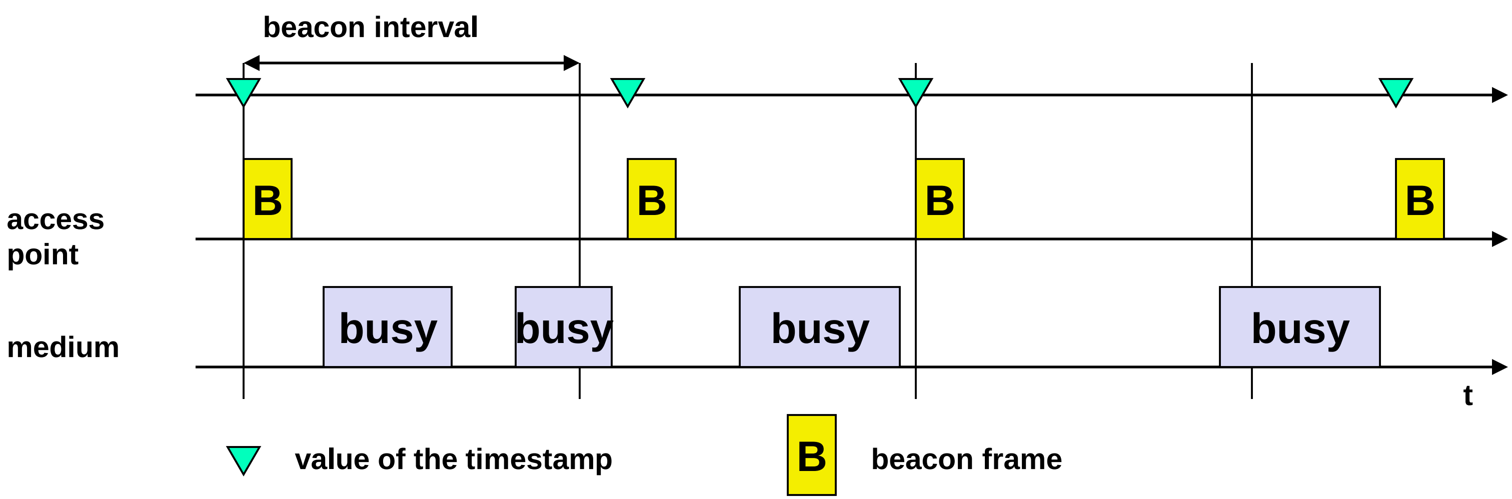
Questions to be answered:

- How does beaconing work in ad-hoc mode?
- How are collisions among beacons avoided in ad-hoc mode?
- What´s the idea behind energy saving in 802.11?
- Does power saving work in ad-hoc mode as well?
- What information does a TIM contain? What a DTIM?
- How does a handover work in 802.11? Why is it so slow, compared to GSM or UMTS?

802.11 – MAC Management

- Synchronization
  - try to find a LAN, try to stay within a LAN
  - synchronization of internal clocks to coordinate access (e.g. SIFS, PIFS, etc.), send beacons, etc.
- Power management
  - sleep-mode without missing a message
  - periodic sleep, frame buffering, traffic measurements
- Association/Reassociation
  - integration into a LAN
  - roaming, i.e. change networks by changing access points
  - scanning, i.e. active search for a network
- MIB - Management Information Base
  - managing, read, write

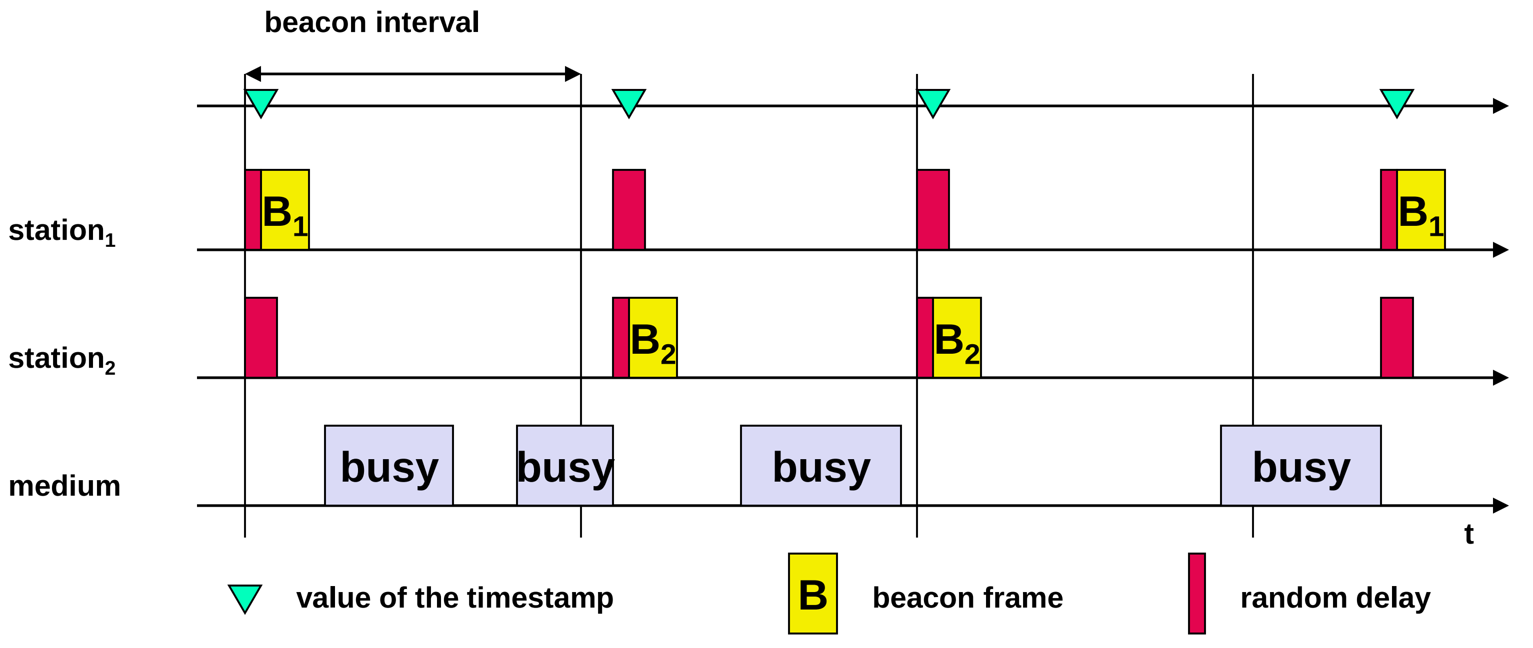
Synchronization Using a Beacon (Infrastructure)



Beacon sent only by access point

- Beacon contains a timestamp and other management information used for power management and roaming
- Beacon may be delayed due to busy medium; beacon interval is not influenced by this!

Synchronization Using a Beacon (Ad-hoc)



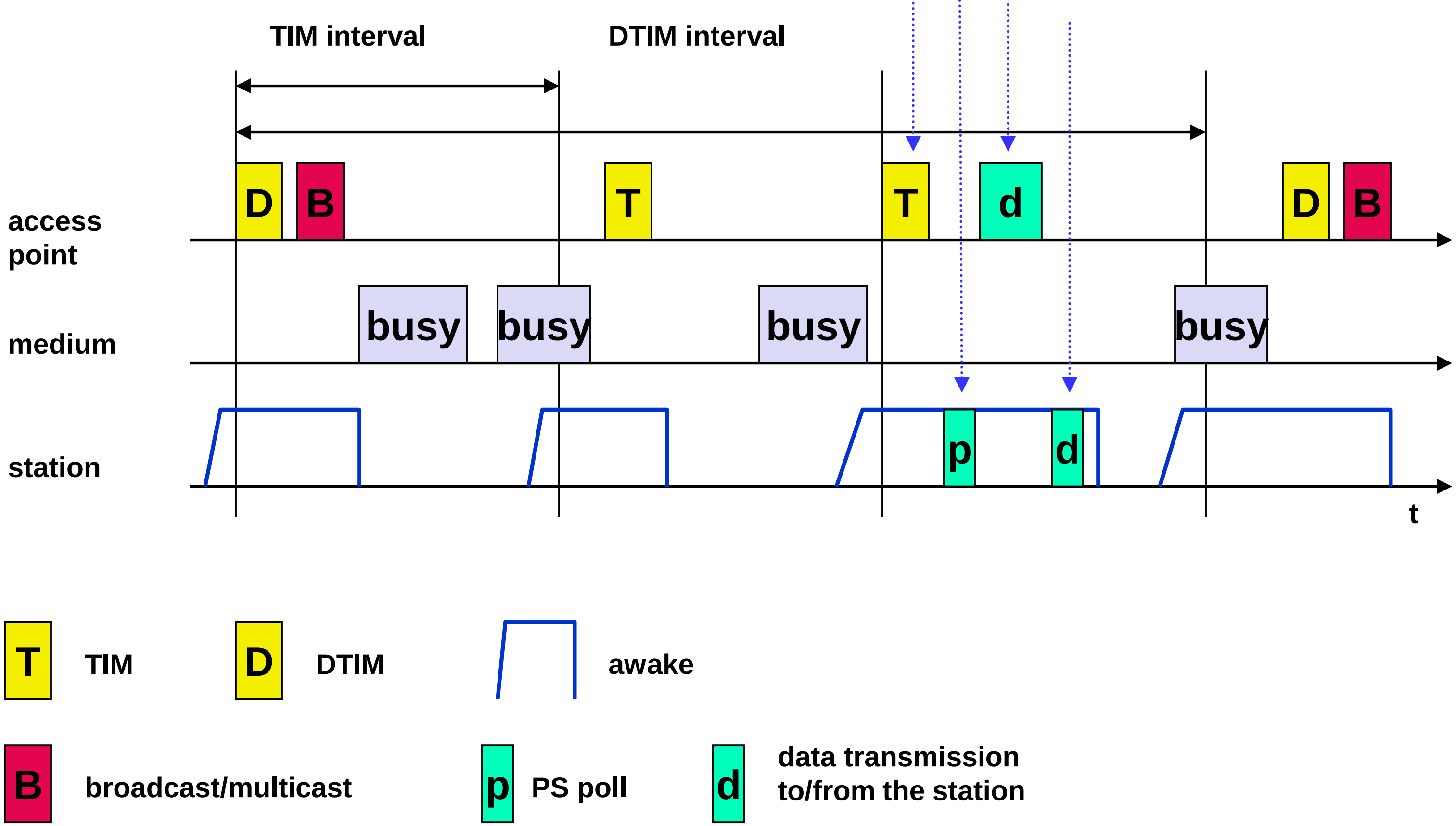
Beacon may be sent by any station (sending of beacon employs a random delay to avoid collisions)

Power Management

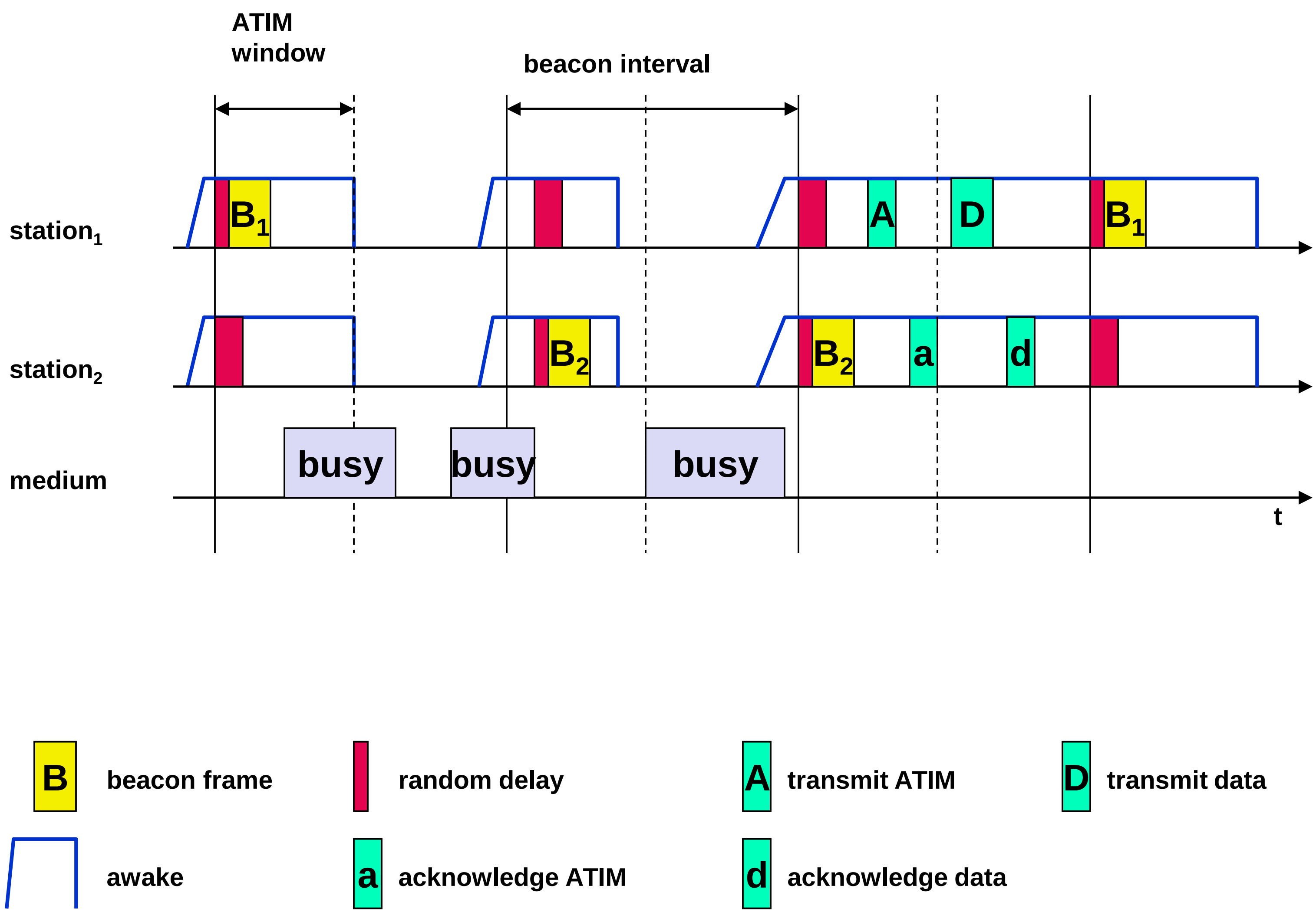
- Idea: switch the transceiver off if not needed
- States of a station: sleep and awake
- Timing Synchronization Function (TSF)
  - stations wake up at the same time
- Infrastructure
  - AP stores frames intended for sleeping stations
  - AP transmits indication about stored frames in periodic beacons (Indication Maps) sent during awake interval
    - Traffic Indication Map (TIM): List of unicast receivers
    - Delivery Traffic Indication Map (DTIM): List of broadcast/multicast receivers
- Ad-hoc
  - Ad-hoc Traffic Indication Map (ATIM)
    - announcement of receivers by stations buffering frames
    - more complicated - no central AP
    - collision of ATIMs possible (scalability?)

Power Saving with Wake-up Patterns (Infrastructure)

- beacon indicates station that data are available
  - station replies with PS (power save) poll and continues listening to the medium
    - AP transmits data
      - station acknowledges the data



Power Saving with Wake-up Patterns (Ad-hoc)



802.11 – Roaming

- No or extremely poor connection? Then perform:
- Scanning
  - scan the environment, i.e.
    - ❑ listen into the medium for beacon signals or
    - ❑ send probes into the medium and wait for answers

  => time-consuming (scan all channels)
- Association Request
  - station sends a request to an AP
- Association Response
  - success: AP has answered, station can now participate
  - failure: continue scanning
- AP accepts Reassociation Request
  - signal the new station to the distribution system
  - the distribution system updates its data base (i.e. location information)
  - the distribution system may inform the old AP so it can release resources
  - 802.11f standard (IAPP – Inter Access Point Protocol)

Physical Layer Variants and Extensions

Questions to be answered:

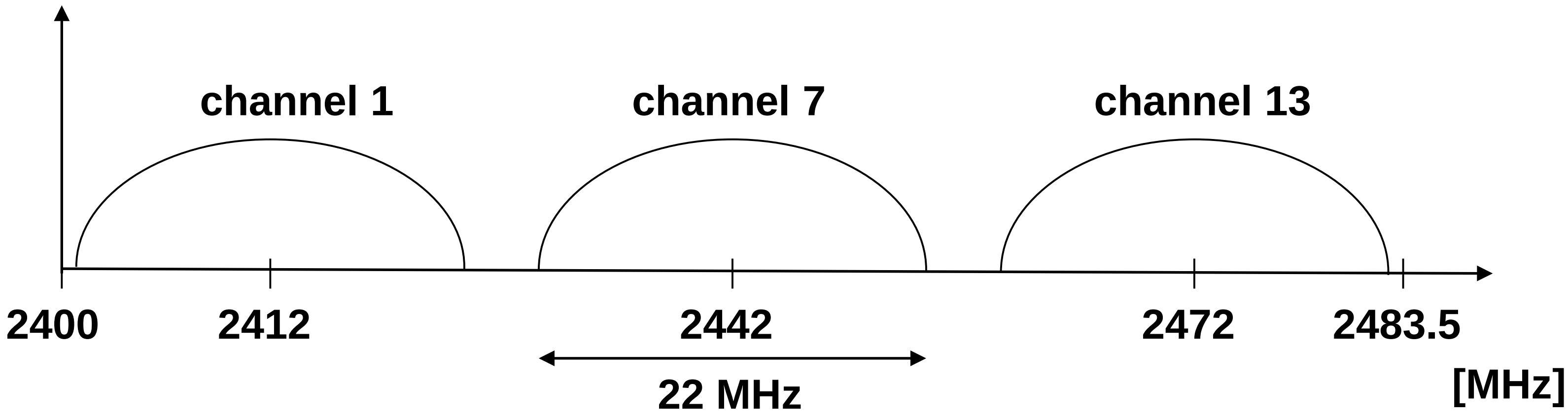
- What are advantages and disadvantages of using higher frequencies?
- Why is the transmit power limited to 100 mW?

WLAN: IEEE 802.11b

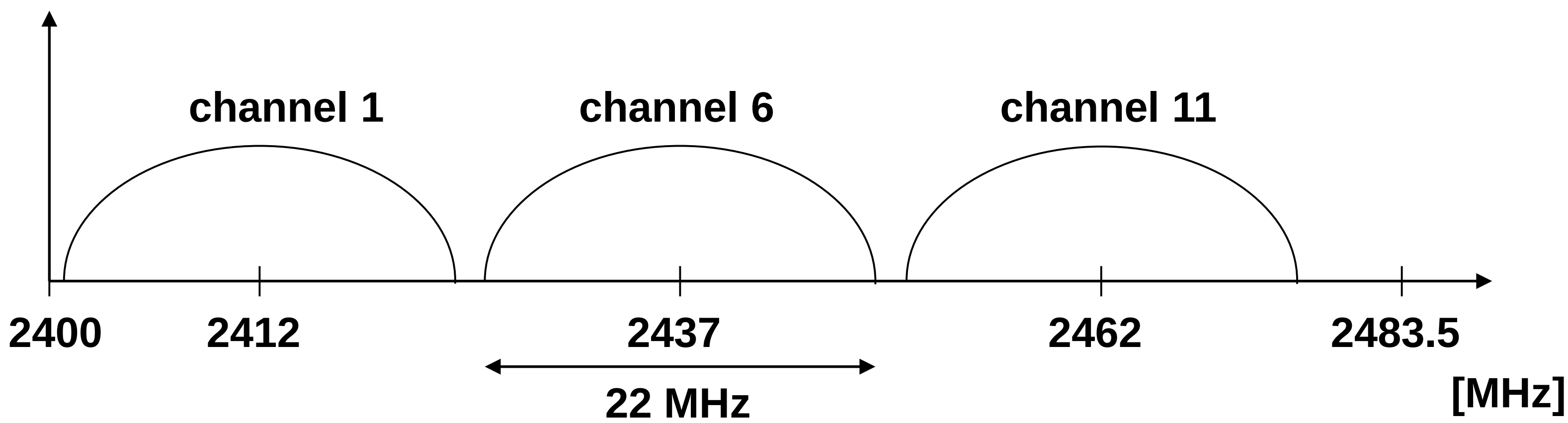
- Data rate
  - 1, 2, 5.5, 11 Mbit/s, depending on SNR
  - User data rate max. approx. 6 Mbit/s
- Transmission range
  - 300m outdoor, 30m indoor
  - Max. data rate ~10m indoor
- Frequency
  - Free 2.4 GHz ISM-band
- Security
  - Limited, WEP insecure, SSID
- Connection set-up time
  - Connectionless/always on
- Quality of Service
  - Typ. best effort, no guarantees (unless polling is used, limited support in products)
- Manageability
  - Limited (no automated key distribution, sym. encryption)
- Advantages/Disadvantages
  - Advantage: many installed systems, lot of experience, available worldwide, free ISM band, many vendors, integrated in laptops, simple system
  - Disadvantage: heavy interference on ISM band, no service guarantees, slow relative speed only

Channel Selection (Non-overlapping)

Europe (ETSI)



US (FCC)/Canada (IC)



WLAN: IEEE 802.11g

- Data rate
  - 6, 9, 12, 18, 24, 36, 48, 54 Mbit/s, depending on SNR
  - User throughput (1500 byte packets): 5.3 (6), 18 (24), 24 (36), 32 (54)
  - 6, 12, 24 Mbit/s mandatory
- Transmission range
  - 150m outdoor, 20m indoor

    54 Mbit/s up to 6 m
- Frequency
  - 2.412~2.472GHz (Europe ETSI)
  - 2.457~2.462GHz (Spain)
  - 2.457~2.472GHz (France)
- Modulation and coding
  - BPSK, QPSK, 16 QAM, 64 QAM
  - 1/2, 2/3 & 3/4 coding
- Security
  - Limited, WEP insecure, SSID
- Connection set-up time
  - Connectionless/always on
- Quality of Service
  - Typ. best effort, no guarantees (same as all 802.11 products)
- Manageability
  - Limited (no automated key distribution, sym. encryption)
- Advantages/Disadvantages
  - Advantage: free ISM band, compatible with 802.11b standard
  - Disadvantage: heavy interference on ISM band, no service guarantees

WLAN: IEEE 802.11a

- Data rate
  - 6, 9, 12, 18, 24, 36, 48, 54 Mbit/s, depending on SNR
  - User throughput (1500 byte packets): 5.3 (6), 18 (24), 24 (36), 32 (54)
  - 6, 12, 24 Mbit/s mandatory
- Transmission range
  - 100m outdoor, 10m indoor

    e.g. 54 Mbit/s up to 5 m, 48 up to 12 m, 36 up to 25 m, 24 up to 30m, 18 up to 40 m, 12 up to 60 m
- Frequency
  - Free 5.15-5.25, 5.25-5.35, 5.725-5.825 GHz ISM-band
- Modulation and coding
  - BPSK, QPSK, 16 QAM, 64 QAM
  - 1/2, 2/3 & 3/4 coding
- Security
  - Limited, WEP insecure, SSID
- Connection set-up time
  - Connectionless/always on
- Quality of Service
  - Typ. best effort, no guarantees (same as all 802.11 products)
- Manageability
  - Limited (no automated key distribution, sym. encryption)
- Advantages/Disadvantages
  - Advantage: fits into 802.x standards, free ISM band, available, simple system, uses less crowded 5 GHz band
  - Disadvantage: stronger shadowing due to higher frequency, no service guarantees

WLAN: IEEE 802.11a

Modulation and coding scheme details

Modulation	Coded bits per sub-carrier	Coded bits per OFDM symbol	Coding rate	Data bits per OFDM symbol	Data rate for 20MHz channel
BPSK	1	48	1/2	24	6 Mbps
BPSK	1	48	3/4	36	9 Mbps
QPSK	2	96	1/2	48	12 Mbps
QPSK	2	96	3/4	72	18 Mbps
16-QAM	4	192	1/2	96	24 Mbps
16-QAM	4	192	3/4	144	36 Mbps
64-QAM	6	288	2/3	192	48 Mbps
64-QAM	6	288	3/4	216	54 Mbps

WLAN: IEEE 802.11h – Regulatory Details

- Spectrum Managed 802.11a
  - power control
  - dyn. channel/frequency selection

- 4 frequency bands:

  - ❑ 5.150 - 5.250 GHz
    - 4 usable channels (100 MHz)
    - indoor only
    - max. 30mW EIRP (.11a)
    - TPC (Transmit Power Control) max. 60mW EIRP
    - combined TPC and DCS/DFS (Dynamic Channel/Frequency Selection) max. 200mW EIRP
    - Turbo Mode: combination of two carriers to reach 108 Mbps

  - ❑ 5.250 - 5.350 GHz
    - 4 usable channels
    - TPC, DCS/DFS mandatory

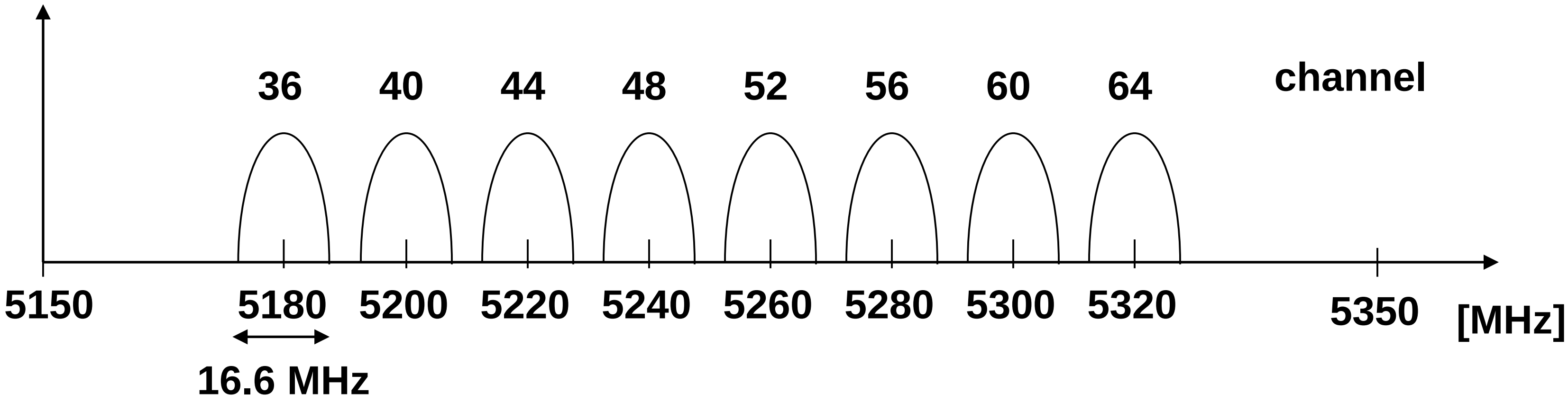
  - ❑ 5.470 – 5.725 GHz
    - indoor and outdoor
    - max. 1W EIRP
    - disallowed in US
    - not supported by all chipsets

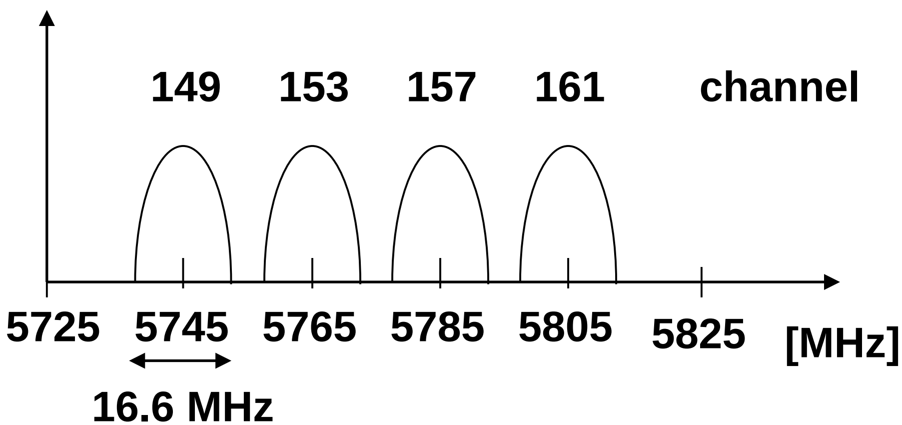
  - ❑ 5.725 bis 5.825 GHz
    - disallowed in Germany

EIRP (Equivalent Isotropic Radiated Power)

Usage in Germany according to the German Federal Regulation (Vorschrift der Regulierungsbehörde für das Telekommunikations- und Postwesen, RegTP, 35/2002)

Operating Channels for 802.11a





center frequency =
5000 + 5*channel number [MHz]

WLAN: IEEE 802.11 – Extensions

- 802.11d: Regulatory Domain Update
- 802.11e: MAC Enhancements – QoS
  - Enhance the current 802.11 MAC to expand support for applications with Quality of Service requirements, and in the capabilities and efficiency of the protocol
- 802.11h: Spectrum Managed 802.11a (DCS, TPC)
- 802.11i: Enhanced Security Mechanisms
  - Enhance the current 802.11 MAC to provide improvements in security
- 802.11n: Throughput enhancement to 108-320 Mbps
- …

- See https://en.wikipedia.org/wiki/802.11 and http://standards.ieee.org/getieee802/802.11.html for newest version

IEEE 802.11-2012 Standard

- 2,4 and 5 GHz ISM band
- Data rates up to 600 Mbit/s (formerly 802.11n)
- Up to 14 (2,4-GHz band), or 28 (5-GHz band*) channels with 20 or 40 MHz bandwidth
- Modulation: DSSS and OFDM
- Transmit power (Germany): 100-1000 mW EIRP
  → distance up to 100 m (more with specialized antennas)
- Encryption according to 802.11i (WPA2)
- Higher data rates including MIMO technology according to 802.11n
- Vehicular communications according to 802.11p
- ...

* country-specific

IEEE 802.11-2016 Standard (Dec. 2016)

802.11ac:

- extension of 802.11n with up to 160 MHz bandwidth in 5 GHz band, 256 QAM and Multiuser-MIMO, data rate up to 1,3 Gbps

802.11ad (WiGig):

- Operation in 60 GHz band, up to 7Gbps data rate

802.11af (White-Fi, Super Wi-Fi):

- Use of TV bands (59 up to 790 MHz) employing Cognitive Radio technologies and geolocation data bases

Note: there is an update every four years with newer developments.

For details see https://en.wikipedia.org/wiki/IEEE_802.11